

Directions: Type `anaconda-navigator` in the terminal to open Anaconda Navigator and launch Spyder. Then complete the following tasks. Write a Python script named `fahrToCel.py` to do the following (one script that contains the code for all three questions) and upload it to Gradescope.

Question 1:

- Prompt the user to enter a speed of an object in miles per hour (mph). Convert it to meters per second using the formula $1 \text{ mph} = 0.447 \text{ m/s}$. Display the answer using a formatted print statement with labels and two decimal places – for example `'60.00 mph = 26.82 m/s'`.
- Add code that calculates how far the object would travel in 10 seconds at the speed entered in part (a). Display the answer using a labeled statement with two decimal places. For example: *"In 10 seconds, the object travels 10.20 meters."* Run this to test it. Do not hardcode the speed. It should work for any speed the user enters.
- Now prompt the user again, this time to enter the weight of an object in pounds; convert this to the mass in kilograms using the formula $1 \text{ pound} = 0.4536 \text{ kilograms}$. Display a labeled answer with two decimal places – for example, `'100.00 pounds = 45.36 kg'`. Test this.
- Add code that converts the mass from kilograms to grams using the formula $1 \text{ kilogram} = 1000 \text{ grams}$. Display a labeled answer with two decimal places. Run this to test it. Do not hardcode the weight. It should work for any weight the user enters.
- Prompt the user to enter:
 - a first name
 - a favorite food
 - a hometown

Display the information in a complete sentence. Example: `Alex from Pittsburgh likes pizza.`

>>> This is the checkpoint for this lab. Show your working code to one of the TAs for 2 points. The remainder of the lab is due at the usual time.

Question 2: Copy the file `'gradebook.txt'` from Canvas; open the file inside Spyder's editor to see what's in it: a set of CSV records representing `<name, id, grade>`. Use the following code to write a script to do the parts described below. We'll talk about file processing more later on; for now, this code opens the gradebook file and reads it one line at a time.

```
with open('gradebook.txt', 'r') as f:
    <your code here: part c>
    for line in f:
        # line now contains one line of the text file
        <your code here: parts a, b, c, d>
    <your code here: part c>
```

- Split each line on the comma and set three variables ***name***, ***id***, and ***grade*** to the returned values.
- Re-assign grade with the integer version of the grade. Then print the three values using a formatted print statement with labels.

- c. Set the variables ***total*** and ***count*** to zero before the loop. After the part b code, add the grade into total and increment count by one. After the loop, compute the average grade in a variable named ***avg*** and display it to 2 decimal places with a label.

Question 3:

- a. Assign the text `Assign the string variable sentence as this sentence.` to a variable called ***sentence***. Display it, then display it in all capitals.
- b. Display the values returned by splitting sentence on space.
- c. Display the values returned by splitting sentence on 's'.
- d. Use `find( )` to display the position of 'ten' in sentence. Then use it again to display the position of the second 'ten'.
- e. Use `replace( )` to change all spaces in sentence into tab characters, but assign this to the variable ***tabby*** and display it.

Deliverable: Upload your .py file to Gradescope. Make sure that your name and Andrew id is at the top of each file. Make sure to document the classes and methods as in the Python guide.